

Advanced (Habanero)

- Q1.** What is the mean of the numbers 2, 4, 6, 8, 10?
(A) 4
(B) 5
(C) 6
(D) 8
(E) 10
- Q2.** Which of the following is a measure of central tendency?
(A) Range
(B) Mode
(C) Variance
(D) Standard Deviation
(E) Interquartile Range
- Q3.** A dataset has the numbers 1, 2, 2, 3, 4, 4, 4. What is the mode?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q4.** What is the median of the numbers 1, 3, 5, 7, 9?
(A) 3
(B) 5
(C) 7
(D) 9
(E) 11
- Q5.** A dataset has the numbers 10, 20, 30, 40, 50. If 10 is removed, what happens to the mean?
(A) Increases
(B) Decreases
(C) Remains the same
(D) Cannot be determined
(E) Becomes zero
- Q6.** Which measure of central tendency is most affected by outliers?
(A) Mean
(B) Median
(C) Mode
(D) Range
(E) Variance
- Q7.** A dataset has the numbers 2, 4, 6, 8, 10. What is the difference between the mean and the median?
(A) 0
(B) 1
(C) 2
(D) 4
(E) 6
- Q8.** Which of the following datasets has the largest standard deviation?
(A) 1, 2, 3, 4, 5
(B) 1, 1, 1, 1, 1
(C) 10, 20, 30, 40, 50
(D) 100, 200, 300, 400, 500
(E) 1000, 2000, 3000, 4000, 5000

Open Ended Questions (FRQ)

- Q9.** A bakery sells a total of 240 loaves of bread per day. The number of loaves sold per day for 5 days is 40, 50, 60, 50, 40. Calculate the mean, median, and mode of the number of loaves sold per day.
- Q10.** Compare the measures of central tendency for two datasets: 1, 2, 3, 4, 5 and 10, 20, 30, 40, 50. Explain which dataset has the larger mean, median, and mode.

Chef's Tips

- Tip 1: Encourage students to use calculators to check their calculations.
- Tip 2: Remind students to show their work and explain their reasoning.
- Tip 3: Allow students to use graph paper to help with calculations and visualizations.